Text Data in Business and Economics

Benjamin W. Arold, University of Cambridge

9. Generative AI

Outline

Intro: Generative AI for Text

- **Type of generative AI that produces text**
- AI systems trained to predict the next word given preceding text
- Typically fine-tuned to follow human instructions and generate responses aligned with human preferences
- Based on deep neural networks with billions of parameters
- Built on transformer models (with attention mechanisms, which endogenously assign varying degrees of importance to different words)

Step 1: Pre-Training

- Calculate conditional probability distribution over words given the preceding words, based on its training data (**Next Token Prediction**)
- **Self-Supervised Learning**: Model is fed text fragments; parameters are adjusted to predict continuation
- Terabytes of data (Wikipedia, scientific articles, books, etc.)
- Neural nets learn language structure:
  - Syntactic structures
  - Relationships between words and concepts they represent
  - Context of sentences and word interactions
  - Relationships between different sentences

Step 2: Instruction Fine-Tuning

- Improves model to follow human instructions
- Example: Pretrained model might predict a continuation of "What is your name?" as "And how old are you?"; this is not what we want
- Instruction fine-tuning trains the model to respond appropriately
- **Supervised learning**: Feeding the model millions of examples for responding to thousands of different instructions
- Used for tasks like summarization, answering questions, brainstorming, etc.

Step 3: Reinforcement Learning

- Improves model by incorporating human feedback
- Human raters provide feedback on different responses
- Helps align model with human preferences, particularly in areas hard to define via instruction fine-tuning
- Example: Penalizing hateful responses
- Noisy process:
  - Partly why LLMs sound authoritative even when hallucinating

Outline

Autoencoding Language Models

- **Main example: BERT**
  - e.g. **BERT = "Bidirectional Encoder Representations from Transformers"**
  - pretrained by dropping/shuffling input tokens and trying to reconstruct the original sequence.
  - usually build bidirectional representations and get access to the full sequence.
  - can be fine-tuned and achieve great results on many tasks, e.g. text classification.

BERT I/II

- Task: Masked language modeling:
  - 15% of words masked
  - if masked: replace with [MASK] 80% of the time, a random token 10% of the time, and left unchanged 10% of the time.
  - model has to predict the original word.
- Architecture:
  - a stack of transformer blocks with a self-attention layer and an MLP.
  - BERT-Large has 24 blocks, embedding dimension of 1024
    $\approx$ 340M parameters to learn.

BERT II/II

- Unlike GPT, BERT attention observes all tokens in the sequence, reads backwards and forwards (bidirectional).
- Corpus:
  - 800M words from English books (modern work, from unpublished authors)
  - 2.5B words of text from English Wikipedia articles
- Various extensions: RoBERTa (Robust BERT); VisualBERT

Outline

Training Large Language Models

BERT

GPT

Generative AI for YOUR Research

Autoregressive Language Models

- **Main example: GPT**
  - e.g. **GPT = "Generative Pre-Trained Transformer"**:
  - pretrained on classic language modeling task: guess the next token having read all the previous ones.
  - during training, attention heads only view previous tokens, not subsequent tokens.
  - ideal for text generation.

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  - make model much bigger

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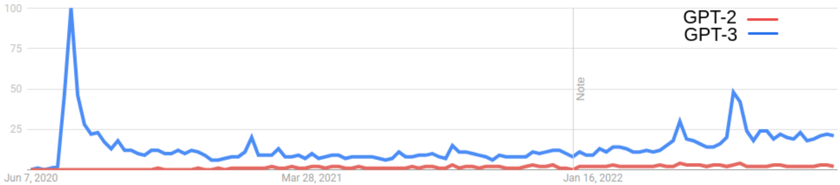
- GPT-3 (2020):
  - use an even bigger corpus (Common Crawl, WebText2, Books1, Books2 and Wikipedia)
  - make model much, much bigger

OPENAI'S NEW MULTITALENTED AI WRITES, TRANSLATES, AND SLANDERS

A step forward in AI text-generation that also spells trouble

By James Vincent | Feb 14, 2019, 12:00pm EST

Howard, co-founder of Fast.AI agrees. "I've been trying to warn people about this for a while," he says. "We have the technology to totally fill Twitter, email, and the web up with reasonable-sounding, context-appropriate prose, which would drown out all other speech and be impossible to filter."



The Washington Post
Democracy Dies in Darkness

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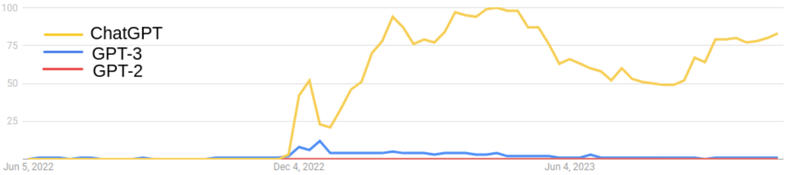
Is ChatGPT the Start of the AI Revolution?

Analysis by The Editors | Bloomberg
December 9, 2022 at 1:39 p.m. EST

The New Chatbots Could Change the World. Can You Trust Them?

Siri, Google Search, online marketing and your child's homework will never be the same. Then there's the misinformation problem.



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- GPT-4 (2023):
  - Multimodal model: can take also images as inputs
  - Trained in two stages:
    1. token prediction (like other GPT models)
    2. **reinforcement learning with human feedback**
  - Much, much, much bigger model

GPT-4o: Innovations and Improvements

GPT-4o (May 2024) introduced the first native multimodal capabilities, improving speed, multilingual support, and real-time processing.

- Multimodal processing of text, images, and audio
- Twice as fast as GPT-4 with lower latency (320ms response time)
- Extended support for over 50 languages
- Enhanced audio processing, including speaker recognition and emotion detection
- Improved visual understanding with better OCR and consistent object rendering
- Real-time API enabling interactive applications with speech capabilities

GPT 5.1: Innovations and Improvements

GPT 5.1 (Dec 2025) delivers substantial gains in reliability, reasoning, and multimodal capability compared to GPT 4.x and 5.0.

- Markedly lower hallucination rates through improved training data filtering and new consistency objectives
- Much stronger long-context performance with more stable memory across long conversations
- Improved mathematical and logical reasoning, especially in multi-step tasks and code generation
- More accurate interpretation of tone, intent, and pragmatics in complex or ambiguous text
- Significantly better multimodal alignment, allowing tighter integration of text, images, and audio
- Faster decoding and lower latency via optimized inference kernels and better parallelization
- Enhanced tool-use reliability, including more accurate structured outputs and API calls

Outline

Choose your GenAI model (updated Dec 2025)

LLM	Price	Tokens[1]	Data Cutoff	Images	PDFs upload
GPT-5.1 (OpenAI)	free / 20$ (Plus) / 200$ (Pro)	128k (ChatGPT) / 400k (API)	09/2024	yes	yes
Claude Sonnet 4.5 (Anthropic)	free / 20$ (Pro)	200k (1M beta)	01/2025	yes	yes
Gemini 2.5 Pro (Google)	free / 19.99$ (AI Pro)	Pro: 1,048,576 ($\approx$1M)	01/2025	yes	yes
DeepSeek-V3.2 (Chat/R1)	free (web/app) + API	128k	not stated	no	yes
Llama 3.1-405B (Meta)	open weights (self-host)	128k	12/2023	no	no (needs tooling)

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- Beyond raw model quality, choose based on ecosystem: connectors (built-in links to your tools and data (e.g., GitHub)), coding agents, file tooling (OCR, pdf upload etc.), admin/security controls.
- Open-weight models (e.g., Llama) can be great for on-prem + cost control, but you must build the "chat app" features (uploads, RAG (Retrieval-Augmented Generation: letting the model answer from your documents reliably), UI).

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Generative AI for Economic Research

- Korinek (2025)[2]:
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- Main takeaway on LLMs for research:

  "employ AI agents in research is to treat them like a PI would treat a team of research assistants: they require careful planning of what is to be done, oversight during execution, and detailed vetting of the final results."

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- Good practices:
  - Provide context (e.g., specify point of view, audience, etc.)
  - Iterate
  - Be patient

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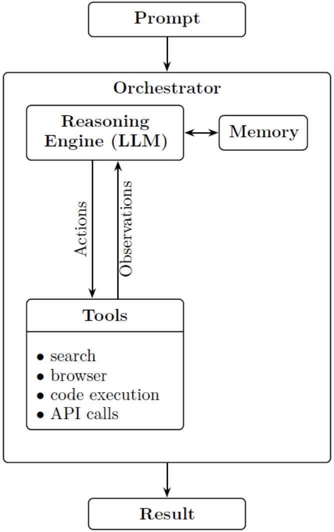
Three paradigms: Chatbots → Reasoners → Agents

- **Traditional LLMs** (broadly available since Nov 2022): fast, fluent, single-pass responses.
- **Reasoning models** (introduced Sep 2024): step-by-step analysis for harder math/logic/coding.
- **Agentic chatbots** (emerged Dec 2024): language + reasoning + *autonomous use of external tools*.
- Mid-2025: leading chatbots can auto-select modes, with an option to "think longer" when needed.

What is an AI agent? (Working definition)

- An AI agent combines:
  - a large language model (LLM),
  - planning (decide steps),
  - memory (track intermediate results),
  - tool access (web search, code execution, APIs, files).
- Unlike a standard chatbot, an agent can:
  - decompose a question into subtasks,
  - gather information from multiple sources,
  - execute code and iterate based on intermediate findings,
  - synthesize results into coherent narratives.

AI Agent Architecture



Comparison of LLM model types

Table 1
Primary Usefulness of LLM Model Types for Research Tasks

Research Category	Traditional LLMs	Reasoning Models	Agentic Chatbots
Ideation & Feedback	Good for initial brainstorming.	**Best for structured feedback and identifying logical flaws.**	Actively scans literature for novelty and grounding.
Writing	**Excellent for drafting, summarizing, and rephrasing existing text.**	Ensures logical flow in complex arguments.	Incorporates real-time web information.
Background Research	*Shortcoming: No live web access.*	Synthesizes info from provided texts.	**Best for live web searches and up-to-date literature reviews.**
Coding	Generates basic code snippets.	**Excellent for writing and debugging complex algorithms.**	**Executes code, tests hypotheses, and interacts with data files.**
Data Analysis	*Shortcoming: Cannot execute code.*	Helps interpret data and suggest approaches.	**Best for end-to-end analysis: data cleaning, coding, visualization.**
Math	*Shortcoming: Unreliable for complex math.*	**Solves multi-step problems and formal proofs at PhD level.**	Can leverage external computational tools to ensure accuracy.
Promoting Research	Drafts initial promotional content.	Tailors summaries for specific audiences.	**Automates creating summaries and posts for multiple platforms.**

Note: The most useful model type in each research category is bolded.

Deep Research agents: multi-agent literature review at scale

- Core architecture:
  - **Lead/orchestrator** plans the strategy and decomposes the question into subtasks.
  - **Specialized subagents** investigate threads in parallel (each with its own context/tools).
  - **Synthesis** integrates findings into a cited report.
- Strength: compresses time-intensive information gathering + initial synthesis dramatically.
- Important caveats:
  - Often compiles existing knowledge (not genuinely novel insight),
  - Can struggle to identify the most impactful frontier papers,
  - Steering depth/breadth is constrained by provider cost limits and usage caps.

Coding agents & “vibe coding”

- Coding agents: terminal-based assistants that can generate, modify, test, and debug multi-file projects.
- “Vibe coding”: building software end-to-end from natural-language prompts.
- Example workflow (econometrics):
  - build a small UI for CSV upload,
  - select variables for OLS,
  - compute regression stats (SEs, t-stats, p-values, R^2),
  - plot scatter + regression line,
  - iterate quickly via follow-up prompts (fix missing values, fix distributions, etc.).
- Broader use: even if you hand-code, these tools are strong for code review and bug-finding.

Under the hood: the basic agent loop

- A simple agent can be structured as:
  1. **Think:** decide what information/tool calls are needed,
  2. **Act:** call tools/APIs (e.g., data retrieval),
  3. **Observe:** parse/validate returned outputs,
  4. **Respond:** produce a grounded natural-language answer.
- Key design idea: make intermediate outputs explicit so they can be checked, logged, and debugged.
- Economists: this pattern supports reproducible pipelines (data access $\rightarrow$ analysis $\rightarrow$ reporting).

Limitations

Practical limitations (why human oversight remains essential):

- hallucinations (incl. bogus citations/statistics),
- error propagation / multi-step cascades,
- brittleness to prompt changes (reproducibility issues),
- prompt injection risks when tools/web are involved,
- not yet reliable as independent economic reasoners.

Example: Pairwise Comparisons – Arold et al. (2025)

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- Have GPT validate pairs of clauses:

  "Which of these sentences from a union collective bargaining agreement is more likely to be interpreted as an entitlement, benefit, or amenity for workers? Answer 'Definitely 1', 'Probably 1', 'Probably 2', 'Definitely 2', or 'Neither'. 1. `[sentence 1]`. 2. `[sentence 2]`.

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- Compute the probability that one clause is more pro-worker than all others
- Validate previous classification

Pairwise Comparisons – Arold et al. (2025)

Table 4: LLM Validation of Worker Rights as Favoring Workers

Clause Type	Clause Frequency (%)	Pro-Worker Frequency (%)
Worker Right	**22.9**	**79.7**
Worker Permission	8.4	67.3
Union Right	2.1	65.7
Firm Obligation	24.7	60.7
Manager Right	0.2	58.4
Worker Obligation	20.9	53.1
Manager Obligation	1.7	44.5
Union Permission	2.0	43.2
Worker Prohibition	3.1	41.6
Manager Prohibition	0.1	38.8
Firm Prohibition	1.5	36.5
Firm Permission	3.4	35.5
Manager Permission	0.4	34.3
Union Obligation	7.0	31.3
Union Prohibition	0.6	28.3
Firm Right	0.9	28.2